

DBDOME · TECHNICAL EXPLAINER

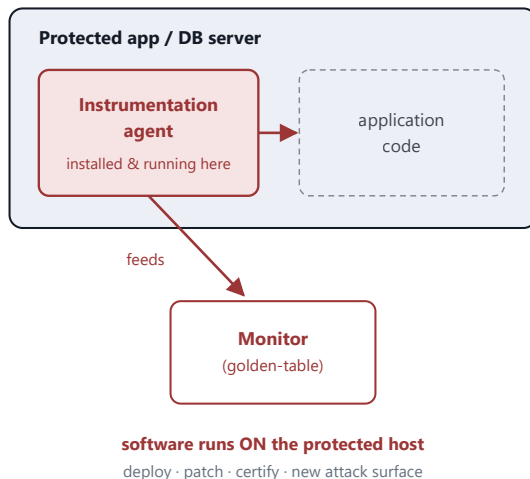
Agentless by Design

Why installing nothing on the protected database is not a convenience — it is the architectural constraint that shapes the whole invention, and the line the closest prior art cannot cross.

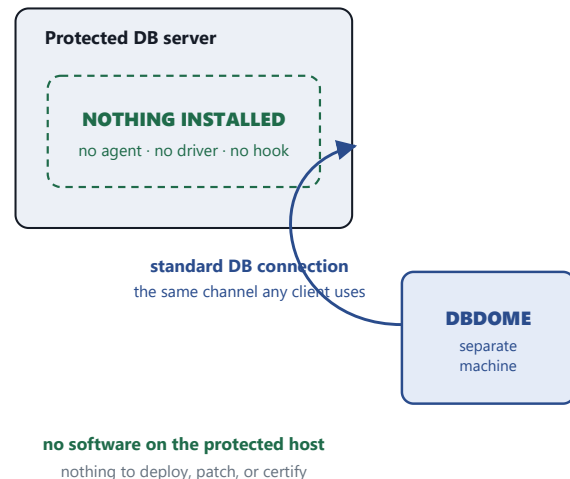
01 The one edge that changes everything

Two ways to watch a database for attacks. They differ by a single component — a piece of software running *inside* the protected server. The prior art requires it. This invention removes it. Everything else follows from that one edge.

AGENT-BASED · prior art (Gupta)



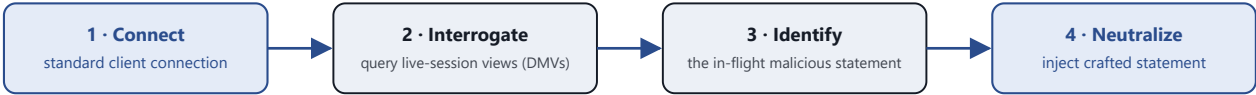
AGENTLESS · DBDOME



Draw the difference and it is a single box. The prior art's method *depends* on the agent inside the protected server — remove it and the monitor has no input. DBDOME's protected server is empty; the defender reaches it only over an ordinary database connection, from a separate machine. That removed box is the whole distinction — operationally, for security, and for the patent.

02 How you watch a server you never enter

If nothing runs inside the database, how does the defender see the attack? It asks the database the same way a monitoring tool would — through the server’s own diagnostic interface, over a normal connection.



every step travels the one connection — the server gains no new software, and no new listening port

Interrogation, not instrumentation. On SQL Server the live-session views (`dm_exec_sessions` , `dm_exec_requests` , `dm_exec_sql_text`) expose the active statement to an ordinary query. DBDOME reads them and responds on the same connection. Nothing is loaded into the server’s process, so there is nothing on the host to be patched, versioned, or turned into a foothold.

03 What the removed box buys

The agent is not a neutral implementation detail. Placing software inside the production database carries costs on every axis that matters to an operator — and removing it reverses each one.

AXIS	AGENT-BASED (PRIOR ART)	AGENTLESS (DBDOME)
Deployment footprint	Install & run code on every protected host	Nothing on the host; one separate monitor
Attack surface added	The agent itself becomes a target inside the trust boundary	No new code, no new port on the server
Blast radius if compromised	Agent runs with in-server privilege — a foothold on the database	Compromise of the monitor never lands code on the DB
Change control / certification	Every agent update must be qualified against the production DB	The protected server is never modified, so nothing to re-certify
Performance impact	In-process hooks consume the server’s own CPU and memory	Work happens off-box; the server just answers queries
Coverage	Only hosts where the agent is installed and current	Any reachable database, immediately

The trade the operator actually cares about: an agent-based defense asks you to put more software onto the very asset you are trying to protect. The agentless design refuses that trade — it protects the database without ever touching it.

04 Why this is the line the prior art cannot cross

THE REJECTION	Application 19/198,051 stands rejected under §103 over Miller in view of Gupta. The combination assumes the two references can be assembled into the claimed method.
THE OBSTACLE	Gupta (Virsec) states, in its published text, that “the instrumentation engine installs and monitors instrumented code on the web server ... or the application server ... executing the web application.” The agent is not optional to Gupta — it is the mechanism by which Gupta correlates requests to queries.
THE CONSEQUENCE	Amending the claim to require agentless interrogation removes the very component Gupta’s method runs on. You cannot modify Gupta to be agentless without destroying its principle of operation — the recognized signature of a combination that teaches away . <i>In re Ratti</i> ; <i>In re Gordon</i> .

Agent-based and agentless are not two settings of one design. They are opposite answers to a single question — *does defensive software run inside the protected server?* The prior art’s answer is yes, by necessity. This invention’s answer is no, by design.

agent-based / prior art agentless defender path the removed footprint

Reference characterization drawn from the published text of US 2016/0337400 (Virsec Systems). SQL Server live-session views named per Microsoft documentation. Explanatory summary prepared for prosecution counsel — the claims define the invention.